controlled single-phase and 3-phase rectifiers using thyristors; DC-to-AC converters; DC-to-DC buck and boost converters; and applications of DIAC, TRIAC, and IGBT. **Co-requisite:** ELEC 432.

ELEC 451	Control Systems	3(3,0)
This course is to explore the modeling of linear dynamic systems via differential equations and transfer functions utilizing state-space and input-output representations; analysis of control systems in the time and frequency domains, study of the classical stability tests, such as the Routh-Hurwitz and Nyquist criteria, and design methods using root-locus plots and Bode plots; and the development of control techniques based on PID, lead and lag networks. **Prerequisite**: CCEE 331.		

ELEC 499	Co-op Work Experience	3(3,0)
This Co-op work experience is designed to provide students with 5 weeks of short-term work in electrical engineering. Students are encouraged to network in the discipline-related industries. A report, a poster, and a PowerPoint presentation are required. **Prerequisites**: ENGL 217; Senior Standing.		

ELEC 533	Advanced Electric Machines and Drives	3(3,0)
This course focuses on the basic structure and principles of DC and AC drive systems. Constant flux and field weakening. Constant torque and constant hp operation. Effect of non-ideal waveform on motor performance. Application of Space Vector to AC machine analysis. Pulse Width Modulation techniques. Multi-level converters and multi-phase motor drives. Modern methods of motor control: Field Oriented Control, Direct Torque Control, etc. Methods of Sensorless Control. **Prerequisite:** ELEC 331 and ELEC 432.		

ELEC 534	Advanced Power Systems	3(3,0)
The course is designed to give students a deeper understanding of power system analysis by covering the following topics: Circuit models, Per-unit representation, Network methods, Load-flow studies, Load-flow control, and Economic dispatch. It also covers the Symmetrical three-phase faults, Unsymmetrical faults, Power system stability, and computer methods. **Prerequisite**: ELEC 330.		

ELEC 595A	BE Summative Learning Project 1	1(1,0)
The first phase of a team-oriented, project-based experience that culminates in the creation of an artifact; milestones include: project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the created device. **Prerequisite**: ENGL 217.		

ELEC 595B	BE Summative Learning Project 2	3(3,0)
The second phase of a team-oriented, project-based experience that culminates in the creation of an artifact; milestones include: project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the created device. **Prerequisite**: ELEC 595A.		

II. Elective Courses

Students may meet their engineering technical electives requirements from the following selection of elective courses.

ELEC 433	High Voltage Engineering	3(3,0)
This course introduces high-voltage engineering, withstand levels, S curves, insulation coordination, and breakdown mechanisms. Non-destructive testing of apparatus: insulation resistance, partial discharge, and HV production measurements for test objects. System over-voltages, switching, lightning, and over-voltage protection devices. **Prerequisite:** ELEC 330.		

ELEC 434	Electrical Power Distribution	3(3,0)
This course covers general considerations; load characteristics; sub-transmission and distribution substations; primary and secondary distribution; secondary network systems; distribution transformers; voltage regulation and the application of capacitors; voltage fluctuations; and protective device coordination. **Prerequisite:** ELEC 330.		

ELEC 444	Electromagnetic Compatibility	3(3,0)
This course introduces Electromagnetic Interference, Conducted and Radiated Emission, Conducted and Radiated Susceptibility, and Product Slippage. EMC standards and Regulations. EMC Theory and Materials Relating to EMC Design. EMC Measurement and test facilities. **Prerequisite:** ELEC 210.		

ELEC 498	Special Topics in Electrical Engineering	3(3,0)
This course is designed to enable students to study current special topics of interest, carefully selected from ELEC-related areas. The contents of such a course are to be determined by the instructor and the department.		

ELEC 503	Artificial Neural Systems	3(3,0)
This course focuses on the foundations of neural network theory and its application in engineering, cognitive science, and artificial intelligence. The course also covers the foundations of machine learning and neural processing algorithms: supervised and unsupervised learning of feed-forward and recurrent neural networks, perception layers, associative memories, and feature maps. **Prerequisite**: MATH 311		

ELEC 504	Lasers and Laser Applications in Engineering	3(3,0)
This course is designed to give students a functional knowledge of lasers. The course covers light, atoms, absorption processes, and spontaneous and stimulated emission of radiation. Moreover, the types and characteristics of lasers are discussed. Laser sources, resonators, and amplifiers are discussed. In addition, applications in engineering, technology, and the biomedical field will be discussed.		

ELEC 531	Electrical Power Systems for Transportation Applications	3(3,0)
Electrical power and energy systems are at the forefront of application development in, for example, electric and hybrid road vehicles, more-electric and all-electric aircraft,		

and traction applications. The associated hardware, technologies, and control methods are crucial to achieving critical global targets in energy efficiency, low-carbon, and low-emissions operations. The greatest challenges arise when we combine new technologies at a large scale and at complex system levels. This course will focus on power systems and control equipment for electric transportation. The course will focus on the equipment needed to support new forms of electric transportation, such as Electric cars, aircraft, and trains. **Prerequisite**: ELEC 432.

ELEC 532	**Application of Power Electronics in Power Systems**	**3(3,0)**
This course focuses on power electronics and its applications in power systems, covering the following topics: Flexible AC transmission systems (FACTS), conventional thyristor-controlled reactors and phase shifters, voltage source inverter (VSI)- based static condenser (STATCON), and unified power flow controller (UPFC). **Prerequisites**: ELEC 330; ELEC 432.		

ELEC 535	**Renewable Energy Systems for the Built Environment**	**3(3,0)**
This course covers Energy and carbon emissions in the built environment, including data on energy consumption in Lebanon. Overview of building services – HVAC, Lighting, HWS, Lifts, and Small Power; data and benchmarks for energy consumption by end use in buildings; methods for estimating predicted energy consumption in buildings; time-based energy demand schedules; the importance of energy efficiency and the Lean-Clean-Green principle. **Prerequisite**: ELEC 210.		

ELEC 536	**Renewable Energy Systems**	**3(3,0)**
This course aims to provide knowledge about energy production from renewable sources, the structure and nature of the interconnected electric power system, and the critical need for environmentally sensitive solutions. In addition, the economic and regulatory policy aspects of electricity and electricity markets are discussed. The course covers the basics of Wind energy conversion systems and microgrids with hybrid power sources. Modeling and control of renewable energy sources such as wind turbine generation, solar panels, fuel cells, and power electronics interfaces will be presented. **Prerequisite**: ELEC 431.		

ELEC 542	**Advanced Power Electronics**	**3(3,0)**
This course is designed to provide students with a functional knowledge of modeling switching power converters, advanced power converter topologies, design constraints, and control methods. It also covers the operation of multi-level DC/AC inverters and matrix converters. **Prerequisite**: ELEC 432.		

ELEC 550	**Advanced Control Systems**	**3(3,0)**
This course focuses on the formulation of the linear control problem using state-space methods, frequency-response and time-response analysis; it introduces students to advanced concepts such as controllability, observability, canonical forms, state transition matrices, stability, nonlinear system linearization, and discrete and time-invariant systems. **Prerequisite**: ELEC 451.		

ELEC 551	Nonlinear Adaptive Control	3(3,0)
This course presents a comprehensive exposition of the theory of nonlinear dynamical systems and their control. It will focus on methods for characterizing and understanding the behavior of systems described by nonlinear ordinary differential equations, as well as methods for designing controllers for such systems. In this course, both classical and modern concepts from nonlinear system theory will be introduced. **Prerequisite**: ELEC 451.		

ELEC 560	Sustainable Energy	3(3,0)
This course is designed to provide you with the methods, tools, and perspectives to understand, critique, and ultimately influence the management of technical, economic, and policy choices regarding energy generation and use. We will focus equally on the technical, economic, political, and environmental impacts of energy. **Prerequisites**: ELEC 431		

ELEC 561	Wind Energy Technology	3(3,0)
This course aims to provide the students with an in-depth understanding of the theoretical, technological, and economic aspects of wind energy systems. Upon successful completion of this course, students will be able to identify, assess, and select the types of wind turbines, estimate the power output of specific wind energy devices and systems, assess the structural suitability of wind towers, and evaluate the key aspects of onshore (urban) and offshore wind energy systems. **Prerequisite**: ELEC 210.		

ELEC 562	Solar Energy Technology	3(3,0)
This course aims to provide students with a systematic understanding of current knowledge, problems, and insights in solar photovoltaic technologies; enable students to evaluate current research and advances in the field; and assess solar PV technologies, develop critiques, and propose solutions. **Prerequisite**: ELEC 210.		

ELEC 597	Advanced Topics in Electrical Engineering	3(3,0)
This course is designed to enable students to study a specific advanced topic of interest in Electrical Engineering. The content outline for such a topic is to be determined by the instructor and approved by the department Chair. **Prerequisite**: Instructor's consent. On demand.		